

KAUSTUBH KALYANRAO SURYAWANSHI

Data Science & Machine Learning Engineer | AI/ML | Production ML & Cloud
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PROFESSIONAL SUMMARY

Data Science & Machine Learning Engineer with hands-on experience building and deploying ML systems for **forecasting and AI analytics**. Skilled in **Python, SQL, scikit-learn, XGBoost, LightGBM, SARIMA, Prophet, FastAPI, Docker, and AWS ECS/Fargate**. Currently completing the **AWS Certified Cloud Practitioner, IBM Data Science Professional Certificate**.

TECHNICAL SKILLS

Programming & Data: Python, SQL, Pandas, NumPy
Machine Learning: Scikit-learn, XGBoost, LightGBM, Classification, Regression, Feature Engineering, Model Evaluation
Time-Series Forecasting: SARIMA, Prophet, Lag Features, Rolling Windows, Seasonality, Forecast Validation, RMSSE, MAE, MAPE
AI & Analytics: NLP, AI/LLM Applications, SHAP, Natural-Language Querying
APIs & Deployment: FastAPI, REST APIs, Streamlit, Docker, OpenAPI
Cloud: AWS ECS/Fargate, EC2, CloudWatch, CI/CD Principles
Databases: PostgreSQL, MongoDB, SQLite
Visualization & Reporting: Matplotlib, Seaborn, Plotly, Excel, Power BI
Developer Tools: Git, GitHub, Jupyter Notebook

SELECTED PROJECTS

DataLens – AI Analytics Platform with Natural-Language Query Interface | GitHub

Python | Pandas | Scikit-learn | XGBoost | LightGBM | FastAPI | Docker | AWS ECS/Fargate

- Architected and deployed an **end-to-end ML analytics platform** with **natural-language querying**, reducing analysis time from 3+ hours to <30 seconds.
- Engineered a 4-stage pipeline for EDA, data-quality checks with PII masking, ML training on 50K+ records, and **SHAP-based explainability** with What-If analysis.
- Deployed with **Docker** and **AWS ECS/Fargate**, achieving **<200ms API latency**; reduced manual analytics workload by 85% and supported 5+ concurrent users.

Walmart DemandLens – Multi-Model Demand Forecasting & Model Selection | GitHub

Python | Pandas | LightGBM | SARIMA | Prophet | Streamlit | FastAPI

- Designed a forecasting framework comparing **SARIMA, Prophet, and LightGBM** across **4 California stores** and **1K+ SKU-store combinations**.
- Engineered lag, rolling-window, calendar, promotion, price, and event features, improving **RMSSE by 15%** versus a naive baseline; selected models using validation performance.
- Built **FastAPI/Streamlit forecasting** interfaces and **28-day rolling forecasts**, reducing generation time from **2 hours to 5 minutes**; reduced stockouts **12%** and **overstock waste 8%**.

Indian Sign Language Detection & Multilingual Translation | GitHub

Python | YOLOv8/YOLOv12 | OpenCV | PyTorch | Ultralytics | PyQt5 | Flask | Gemini API

- Developed a **real-time computer-vision system** using **custom-trained YOLO models** to detect Indian Sign Language gestures from **webcam** video.
- Built **gesture buffering** and **sequence processing** to convert detected signs into text and generate coherent translations with **Gemini API**.
- Implemented **multilingual** output for **English, Hindi, Marathi, Punjabi, and Tamil** with interactive PyQt5/Flask interfaces; included **custom dataset** creation and **YOLO training workflows**.

EDUCATION

B.E. in Information Technology | University of Mumbai | 2026 | CGPA: 7.3/10

ACHIEVEMENTS & RECOGNITION

- 2nd Place, XZIBIT 2026 National Project Competition** — Among **150+ entries**; recognized for production deployment, scalability, and real-world applicability.
- Finalist, 20th Aavishkar National Research Convention (2025)** — Selected among top 15 student research projects; presented patent-pending work on automated food product ingredient analysis.
- Published Patent — SCAN-TO-KNOW:** Automated Ingredient Analyzer for Packaged Foods — Addresses nutrition transparency and consumer safety; filed with Indian Patent Office, supporting scalable food industry compliance.

PROFESSIONAL CERTIFICATIONS

- IBM Data Science Professional Certificate**
- AWS Certified Cloud Practitioner**
- UNIVERSITY OF MICHIGAN Python For Everybody**